Entity and Attributes:

Theatre Table

- theatreId: Unique identifier for the theatre
- title: Name of the theatre
- location: Address or area where the theatre is located
- latitude: Latitude of the theatre
- longitude: Longitude of the theatre

Movie Table

- movieId: Unique identifier for the movie
- title: Name of the movie
- directorId: Reference to personalityId of the director
- heroId: Reference to personalityId of the hero/lead actor
- languageId: Reference to the language table
- duration: Total runtime of the movie

TheatreShow Table

- theatreId: Reference to the theatre where the show is running
- movieId: Reference to the movie being screened
- date: Show date (used to display next 7 available dates)
- startTime: Show start time
- endTime: Show end time
- status: Indicates show availability (active, cancelled, housefull, etc.)

FilmPersonality Table

- personalityId: Unique identifier for a film personality
- name: Name of the director or actor

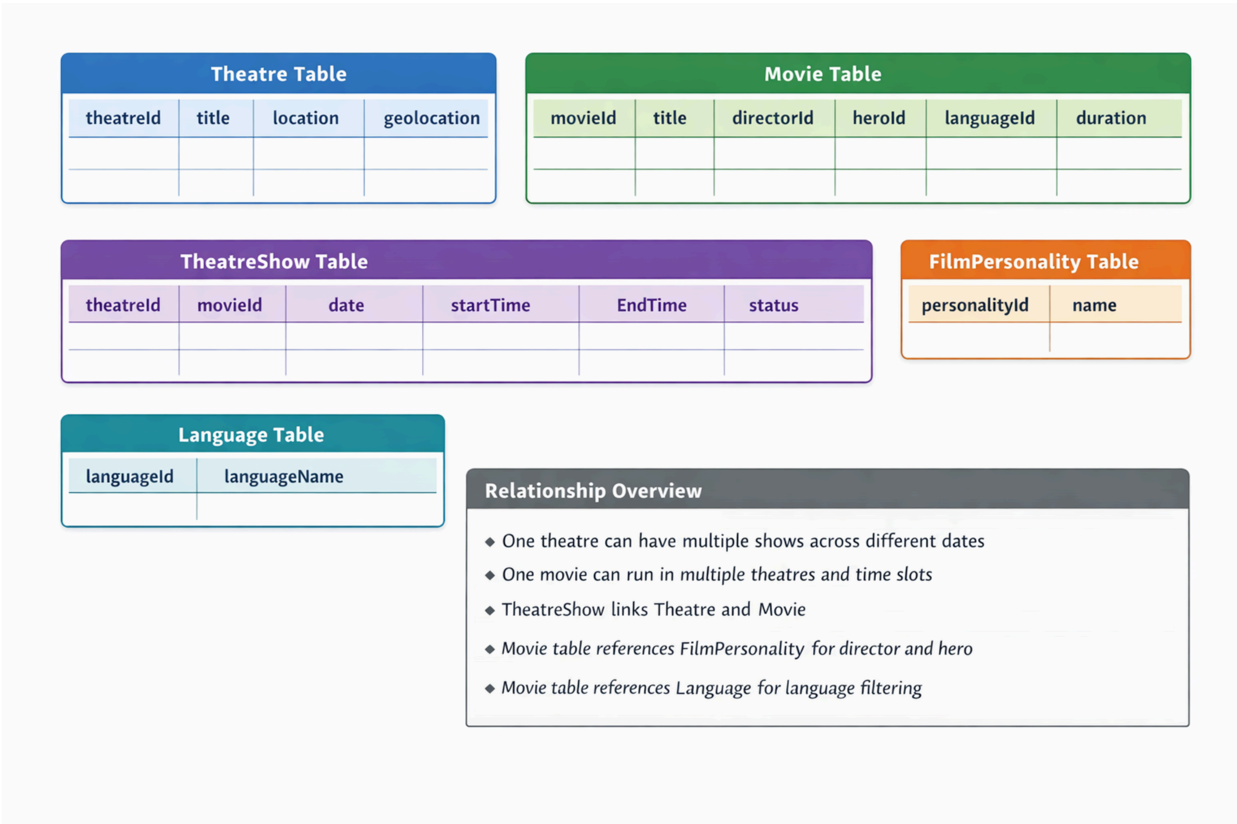
Language Table

- languageId: Unique identifier for the language
- languageName: Name of the language

Relationship Overview

- One theatre can have multiple shows across different dates

- One movie can run in multiple theatres and multiple time slots
- TheatreShow acts as a mapping table between Theatre and Movie
- Movie table references FilmPersonality for director and hero details
- Movie table references Language to support language-based filtering



<u>P1: SQl Queries for creating and inserting data:</u>

1.Theatre Table:

CREATE TABLE Theatre (

theatreId INT AUTO_INCREMENT PRIMARY KEY,

title VARCHAR(255) NOT NULL,

location VARCHAR(255),

latitude DECIMAL(9,6),

longitude DECIMAL(9,6)

);

-- Sample entries

INSERT INTO Theatre (title, location, latitude, longitude) VALUES

('Cineplex Mall', 'MG Road, Bangalore', 12.971598, 77.594566),

('Galaxy Cinema', 'Brigade Road, Bangalore', 12.976231, 77.603287),

('Star Theatre', 'Indiranagar, Bangalore', 12.971891, 77.641151);

2. FilmPersonality Table

CREATE TABLE FilmPersonality (

personalityId INT AUTO_INCREMENT PRIMARY KEY,

name VARCHAR(255) NOT NULL

);

-- Sample entries

INSERT INTO FilmPersonality (name) VALUES

('Christopher Nolan'),

('Robert Downey Jr.'),

('Leonardo DiCaprio'),

('Quentin Tarantino'),

('Scarlett Johansson');

3. Language Table

CREATE TABLE Language (

```
    languageId INT AUTO_INCREMENT PRIMARY KEY,
    languageName VARCHAR(50) NOT NULL
);

-- Sample entries
INSERT INTO Language (languageName) VALUES
('English'),
('Hindi'),
('Tamil'),
('Telugu');
```

4. Movie Table

```
CREATE TABLE Movie (
    movieId INT AUTO_INCREMENT PRIMARY KEY,
    title VARCHAR(255) NOT NULL,
    directorId INT,
    heroId INT,
    languageId INT,
    duration INT, -- in minutes
    FOREIGN KEY (directorId) REFERENCES FilmPersonality(personalityId),
    FOREIGN KEY (heroId) REFERENCES FilmPersonality(personalityId),
    FOREIGN KEY (languageId) REFERENCES Language(languageId)
);
```

-- Sample entries

INSERT INTO Movie (title, directorId, heroId, languageId, duration) VALUES

('Inception', 1, 3, 1, 148),

('Iron Man', 4, 2, 1, 126),

('Pulp Fiction', 4, 3, 1, 154);

5. TheatreShow Table

CREATE TABLE TheatreShow (

theatreId INT,

movieId INT,

date DATE,

startTime TIME,

endTime TIME,

status ENUM('active', 'cancelled', 'housefull') DEFAULT 'active',

PRIMARY KEY (theatreId, movieId),

FOREIGN KEY (theatreId) REFERENCES Theatre(theatreId),

FOREIGN KEY (movieId) REFERENCES Movie(movieId)

);

-- Sample entries

INSERT INTO TheatreShow (theatreId, movieId, date, startTime, endTime, status) VALUES

(1, 1, '2026-01-18', '10:00:00', '12:28:00', 'active'),

(1, 2, '2026-01-18', '13:00:00', '15:06:00', 'housefull'),

(2, 1, '2026-01-18', '11:00:00', '13:28:00', 'active'), (3, 3, '2026-01-18', '18:00:00', '20:34:00', 'active');

P2:Sql query to list down all the shows on a given date at a given theatre along with their respective show timings.

```
SELECT t.title AS theatre_name,
       m.title AS movie_name,
       m.duration,
       ts.startTime,
       ts.endTime,
       ts.status
FROM TheatreShow ts
INNER JOIN Theatre t ON ts.theatreId = t.theatreId
INNER JOIN Movie m ON ts.movieId = m.movieId
WHERE ts.date = '2026-01-18'
  AND t.title like ‘%Cineplex Mall’;
```